

Trevor Nichols

Contact

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01/02/2006
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Career Object

Detail-oriented professional with experience in software development, API endpoint design and implementation, AI/NN training, and frontend design. Deep understanding and passion for backend integration with Microsoft Azure, GitHub actions, and Nix CI/CD. Praised for creative problem-solving, meticulousness, and technical depth.

Skills

6 years	5 years	4 years	< 3 years
- Linux - Docker - Podman - Docker Compose - Systemd - Office Suite	- JS/TS/Node/Deno - Github Actions - Java - Nix/Flakes - HTML/CSS - Python	- MD Docs - QEMU - C#/.Net - Microsoft Server - Ubiquiti Unifi - Docker Swarm	- Azure - Quarto - Rust

Recent Experience

Software Systems Architect Consultant Feb 2025 - Current

Dr. Jolly Bioinformatics Lab | Cleveland, OH, USA

- Built CI pipelines** to build and use specialized bioinformatics software
 - Utilized both Nix and GitHub Actions for artifact building and dev environments
 - Rearchitected and revitalized software to be able to use full advantage of current hardware
 - Identified and built software to enable optimal computing efficiency in data processing pipelines
 - Customized software led to an over 70% increase in compute efficiency
- Managed System Deployments** for Researcher Workstations
 - Resident Expert in Linux providing consultancy for researchers
 - Sped up software usage problem resolution by over 6x
 - Created specialized tools and scripts to streamline research data processing

Senior Researcher - Lead Backend Engineer Sep 2024 - Aug 2025

AI4EDU at Case Western Reserve University | Cleveland, OH, USA

[github://os-computational-economics/ai4edu-api-service](https://github.com/os-computational-economics/ai4edu-api-service)

- Built REST backend services** for our in-house AI chatbot using fastAPI
 - Integrating with multiple APIs including OpenAI, Anthropic, Claude, and our internal offline LLM for private compute
 - Serves 150 people across the economics and business departments at the university to query and synthesize upon internal private data
- Led the backend team**

- Enforced rigorous Python coding standards through **Ruff** with comprehensive lint, security, and type checking
- Regularly performed in-depth code review and constructive feedback to attain high code quality across our repo
- **Built and maintained databases** using PostgreSQL, DynamoDB, and Pinecone
- **Developed workflows and CI/CD** using GitHub workflows
 - Sped up deployment and delivery by over 75% with automated checks and deployment

Open Source Contributor
Aug 2022 - Ongoing

Designed and Created

- **tnichols217/CSDS133-final** - Gerrymandering simulation written in PostgreSQL
- **tnichols217/Bayesian-Belief-Propagation** - Bayesian Graphical Learning library for TS
- **tnichols217/htms** - HTML framework for server-rendered, client side scriptless static sites
- **tnichols217/DSCI453-final-project** - Small NN meta-analysis for AI image detection
- **tnichols217/obsidian-columns** 200+ stars, 40k+ downloads - Column support for the Obsidian notetaking app

- Developed and Contributed to
- **NixOS/nixpkgs** 19.3k stars - The package repository for the Nix package manager and NixOS linux distro
 - **Vinzent03/obsidian-git** 7.3k stars - Git support for the Obsidian notetaking app
 - **beta-nu-theta-chi/bylaws** - designed entire CI pipeline

IT Technician / Programmer Intern
Aug 2022 - Aug 2023

Dalat International School | Tanjung Bungah, Penang, Malaysia

- **Fullstack development** utilizing Aurelia, C#, .NET, Azure functions, and SQL.
- Managed a private network through **Unifi** network management software
- **Maintained and initiated a major refactoring** on an internal management system serving 100+ educators

Education

Case Western Reserve University
Aug 2023 - May 2027

Masters and Bachelors of Science (M.S./B.S.) in Computer Science Candidate

Sec. Bachelors in Data Science, Computer Engineering, Electrical Engineering
Cleveland, OH, US

Senior Status, GPA 3.805	Specialized in
• Questbridge scholar • University merit-based scholar • Dean's High Honors x4	• AI and ML • Data Science Systems and Modelling • Systems Programming • Theoretical Statistics • Hardware Logic Design • Database Systems
• CWRUnix (Case Linux Club) - President • The Fraternity of Theta Chi - Scholarship Chair, Assistant Treasurer, Executive Member at Large since 2024	